R6.6 — Genus 2 under a Nagao-style oracle, with a Richelot image arm

Toy instances, $y^2 = x^5 + b$ over F_p, $p \in \{1051, 2131, 4111\}$; 40 random bases of size 80 per cell; bootstrap 95%. No sub-rho claim.

Verdict: confirmed — the falsifier did not fire.

- A-closed base: unknowns gain 10.00 [10.00, 10.00] on all six curves (falsifier window [8, 12]); class factor at $m = 3$ is 0.95–1.04.
- Richelot image: class factor 3.53–4.20 (R arm) and 3.60–4.07 (literal $\varphi(F)$ on J'); all intervals inside [3, 5].
- Translation closure: $|(C - \infty) \cap (C - \infty - t)| \leq 2$ for every tested t, so no saturated degree-1 base exists.
- Richelot validated: $\#J' = \#J$, 0/300 homomorphism failures, kernel exactly the (2,2) group, index-4 density 0.246–0.251.
- Nagao $m = 2$ oracle: 49–58 counted F_p multiplications per call, 0 disagreements with exact enumeration.

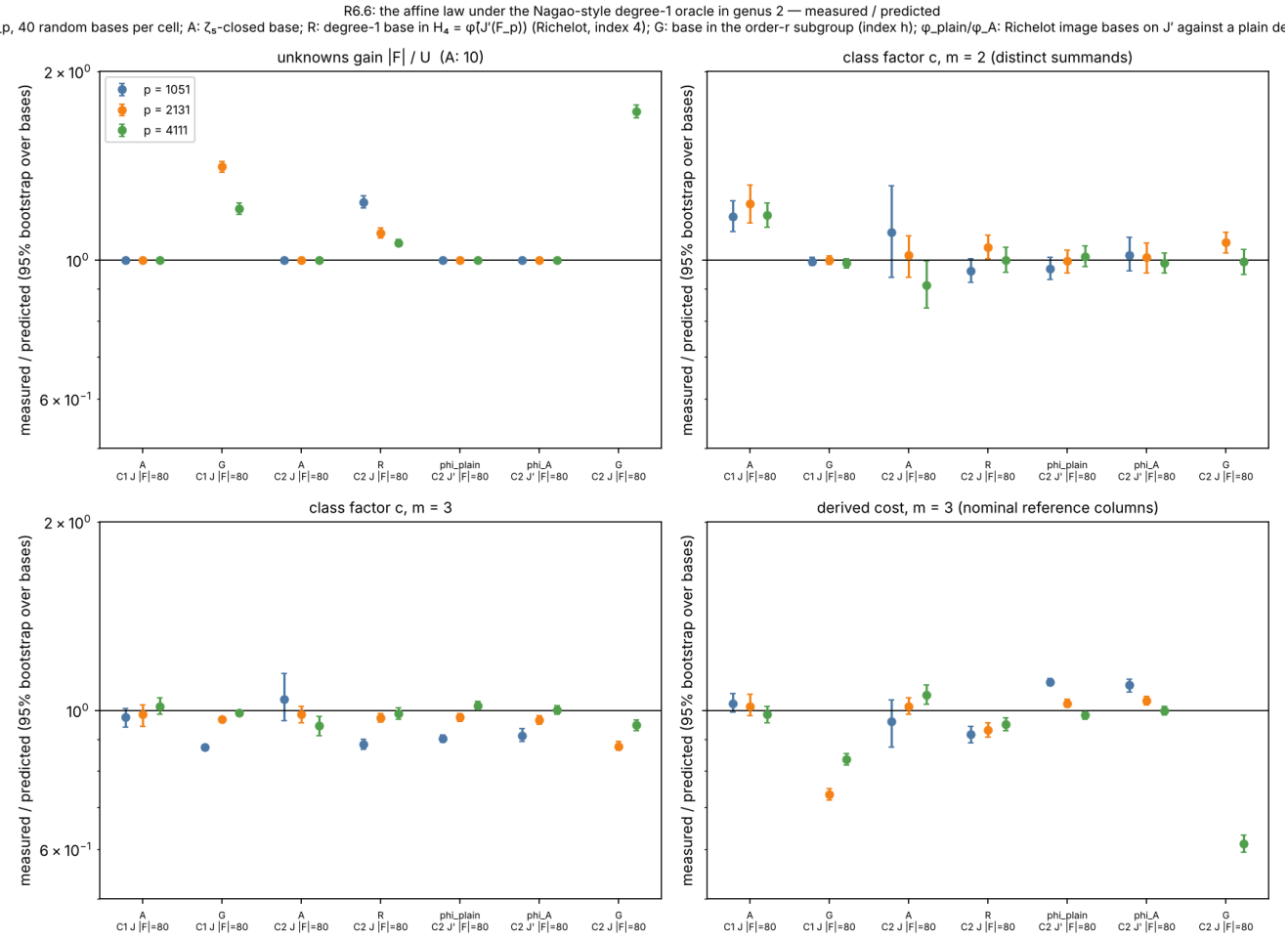


Figure 1: Measured / predicted per arm and p (1 = law).

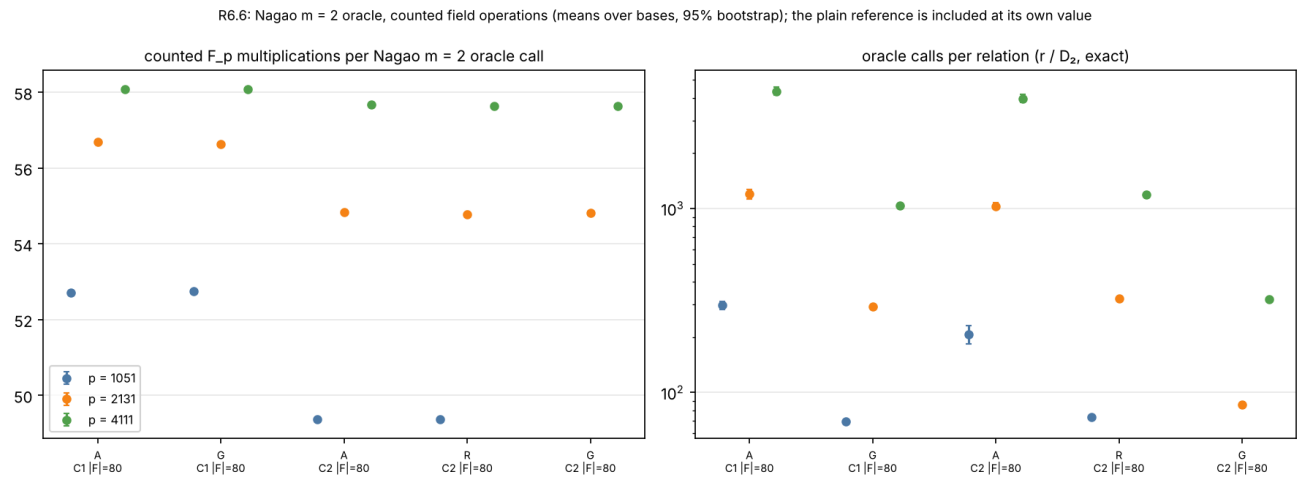


Figure 2: Nagao $m = 2$ oracle: counted multiplications per call and calls per relation.

Details, tables, commands and limits: `RESULTS.md`; raw data: `raw/*.json`.